Two capacitors

X17 (2017) — English translation

In the electrical circuit, the diagram of which is shown in the figure, at first the key is open, there is no charge on the capacitors. The emf of the source is $\mathcal{E} = 10$ В, the resistances of the resistors are the same and equal to $R = 1{,}0$ кОм, the electrical capacitances of the capacitors are the same and equal to $C = 1{,}0$ мкФ. The current source is ideal. The key is closed.

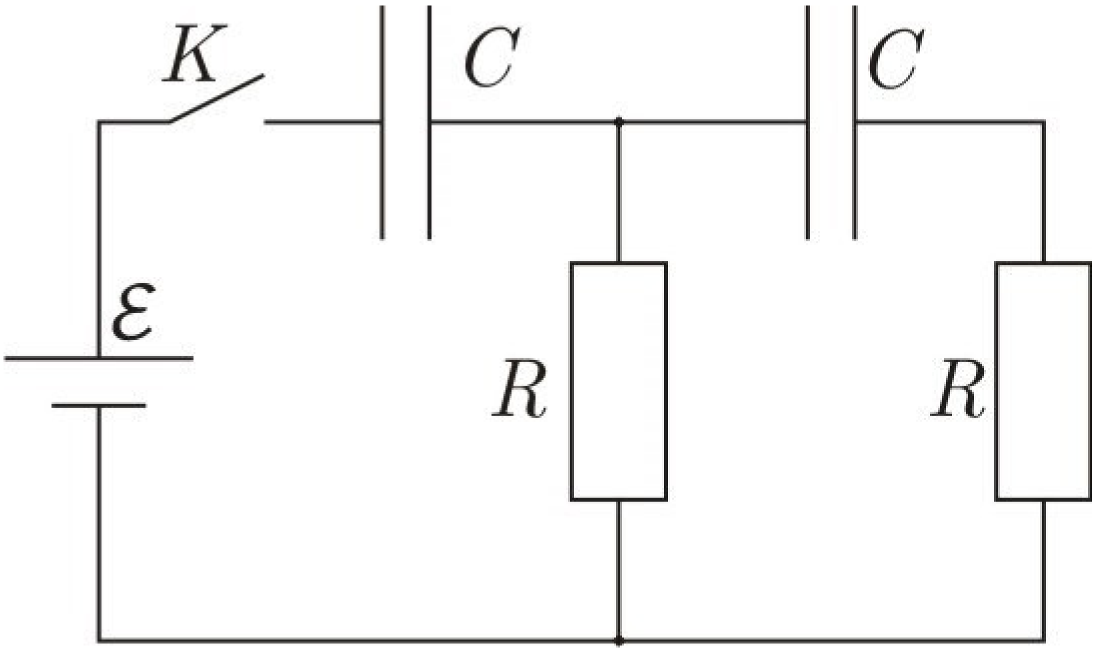

Fig. 1.

A1	What is the charging current through each of the capacitors immediately after the switch is closed? Write down the formulas and give numerical values.	**1.60**
A2	What are the voltages on the capacitors after the time interval $\tau = 1{,}0$ с? Write down the formulas and give numerical values.	**1.60**
A3	Draw on one graph the qualitative dependence of $U(t)$ voltage on time on each of the four elements of the circuit. Indicate on the graph the asymptotes and singular points (zero crossings and extrema) for each of the dependencies.	**4.00**
A4	If for at least two different dependencies any of the special points occur at the same point in time, determine the corresponding values of time and stress. An exact solution is quite a difficult task, so you need to use numerical approximation methods. The permissible error is 10%.	**2.80**

Source: https://pho.rs/p/307